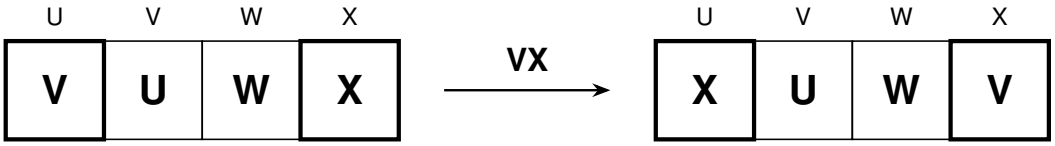
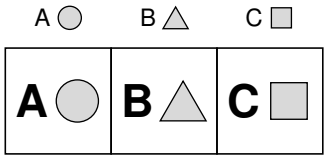
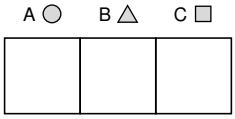
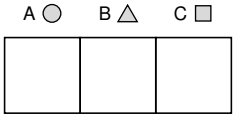
The homes stay in place. A card is at home when it sits under its own letter and picture. A swap trades the places of any two cards, near or far. Write each swap as the letters of the two cards it trades.

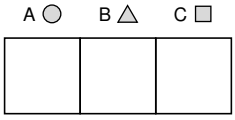


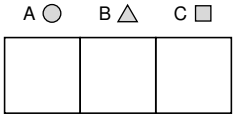
Problem 1: Start with A, B and C at home. Swap cards until no card is at home, using as few swaps as you can.



swaps _____ 

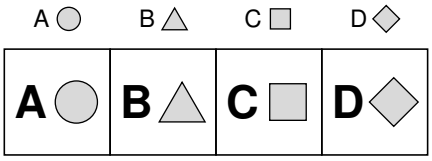
swaps _____ 

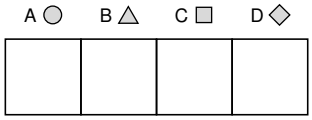
swaps _____ 

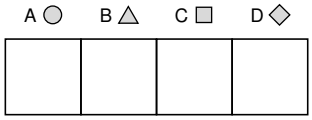
swaps _____ 

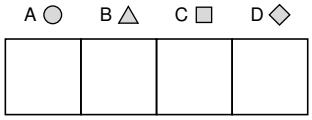
fewest swaps

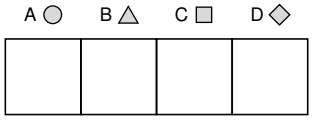
Problem 2: Do Problem 1 again with A, B, C and D. Which rows can you end with in the fewest swaps?

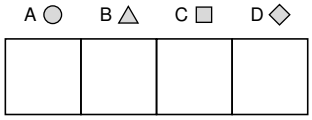


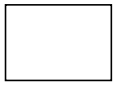
swaps _____ 

swaps _____ 

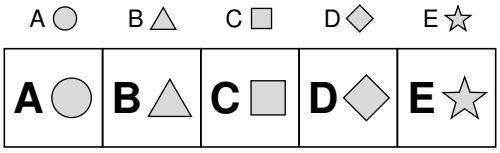
swaps _____ 

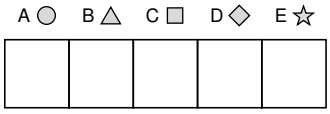
swaps _____ 

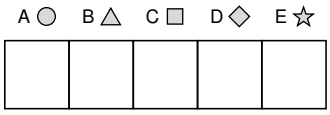
swaps _____ 

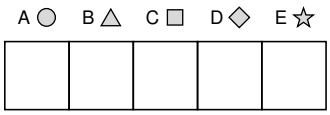
fewest swaps 

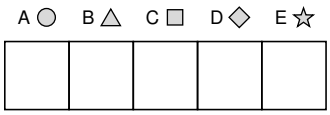
Problem 3: Start with all five cards at home. Can two swaps leave no card at home? Can three?



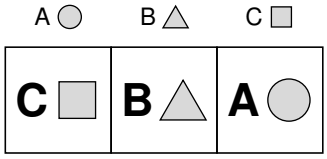
swaps _____ 

swaps _____ 

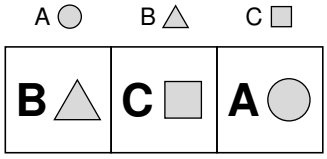
swaps _____ 

swaps _____ 

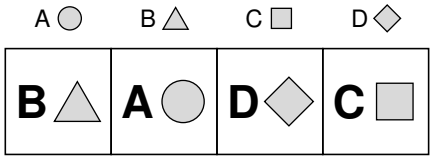
Problem 4: Set out each row. Bring every card home using as few swaps as you can.



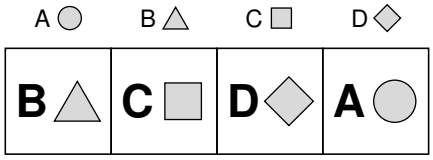
swaps _____ fewest swaps ☐



swaps _____ fewest swaps ☐



swaps _____ fewest swaps ☐



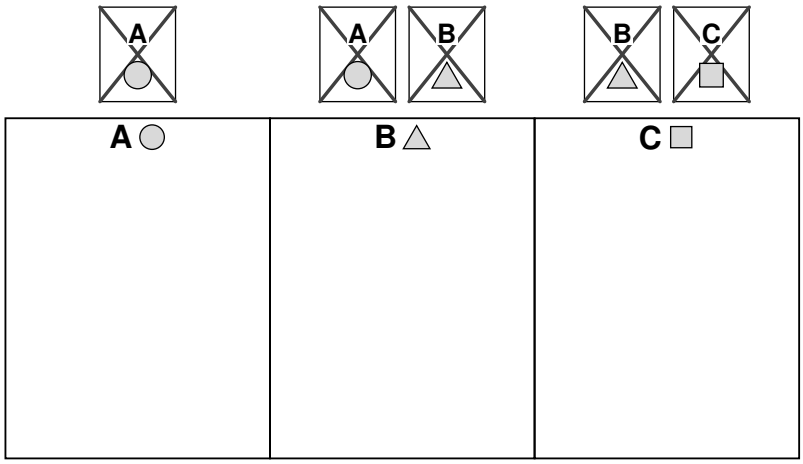
swaps _____ fewest swaps ☐

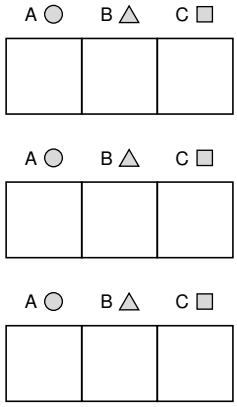
Problem 5: Put a counter on every letter at home on the six A–C outcome cards. Are there more counters, more outcome cards, or the same number, and how many counters does each letter get?

counters ☐ outcome cards ☐

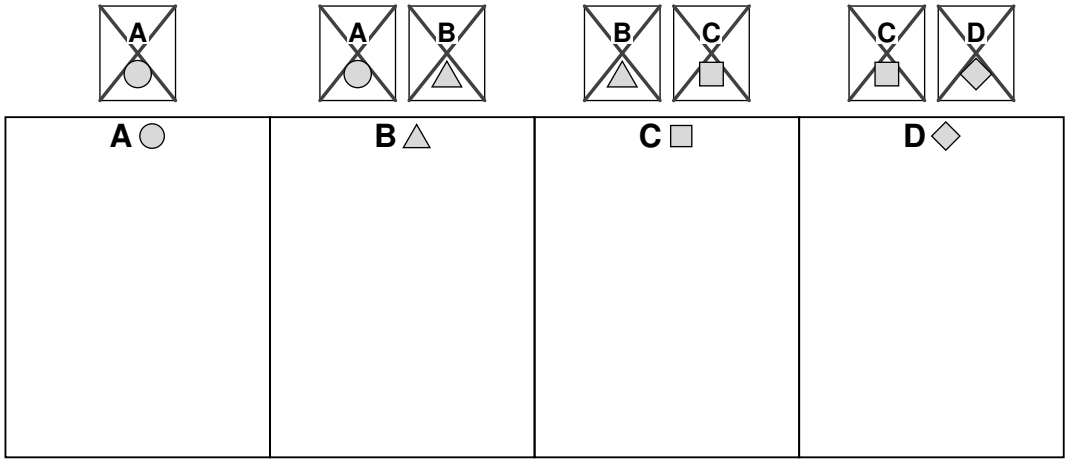
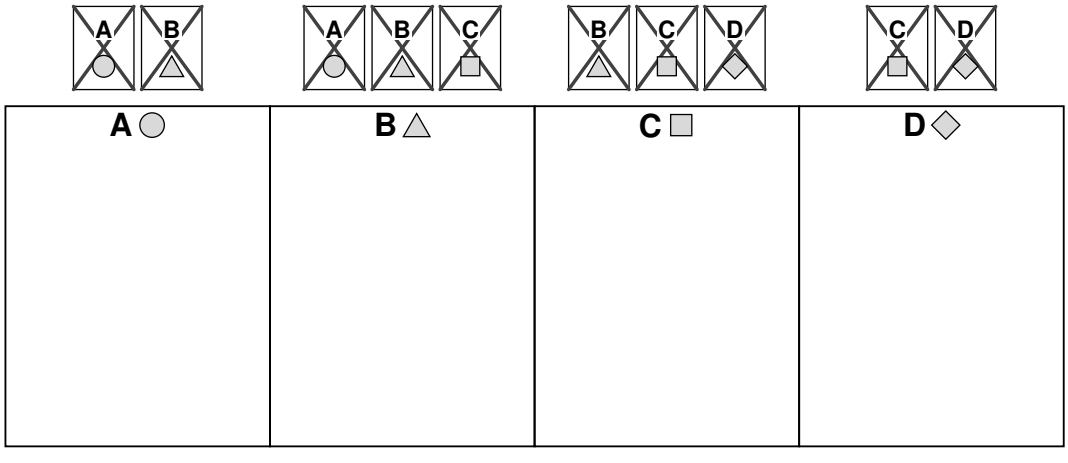
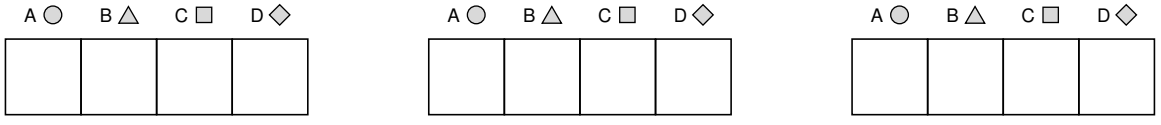
A ○ ☐ B △ ☐ C □ ☐

Problem 6: A card may not go in a home that shows its picture crossed out. Find every way to put A, B and C in these homes.





Problem 7: Find every way to put A, B, C and D in these homes.



Problem 8: Find every way to put A, B, C and D in these homes.